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Noise

Thermal Noise

Noise power input to a device from a matched R source at temperature T Kelvin
B (Hz) is single sided, only positive spectrum, with rectangular window

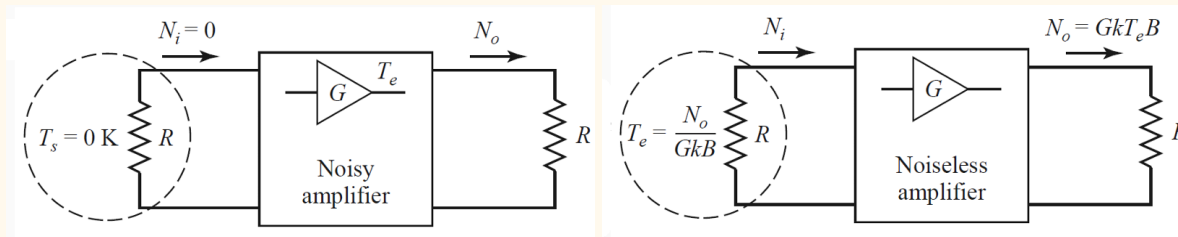
$$N = kTB$$

$$k = \text{Boltzmann's constant} = 1.380649 \times 10^{-23} \text{ J/K}$$

Equivalent temperature for a matched R source to deliver the same (white) noise power to input of device

$$T_e = \frac{N}{kB}$$

Equivalent temperature for amplifier is referred to the input, so must divide the gain



$$T_{e, amp} = \frac{N_{amp}}{kGB}$$

Noise Factor (F)/Noise Figure (NF)

Noise Factor (F)/Noise Figure (NF) of a device is characterized when the input is kT_0B

$$T_0 = 290 \text{ K}$$

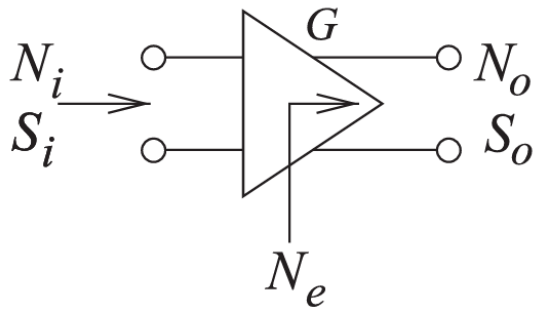
$$SNR = \frac{\text{Signal power}}{\text{Noise power}}$$

$$F = \frac{SNR_{i, T_0}}{SNR_o} = 1 + \frac{N_e}{GN_{i, T_0}} = 1 + \frac{T_e}{T_0} \quad \text{where input noise power } N_{i, T_0} = kT_0B$$

$$NF = SNR_{i, dB} - SNR_{o, dB} \quad \text{where input noise power } N_{i, T_0} = kT_0B$$

$$NF = 10 \log(F) \quad F = 10^{NF/10} \quad G_{dB} = 10 \log(G) \quad G = 10^{G_{dB}/10}$$

$$F \geq 1, NF \geq 0 \text{ dB}, T_{e, amp} \geq 0$$



This is true regardless, N_e being the excess noise added by the device

$$N_o = GN_i + N_e$$

$$N_o = GkT_i B + (F - 1)GkT_0 B = GkB(T_i + T_e) = GkB(T_i + T_{cas})$$

Excess noise added can be expressed as the equivalent noise temperature of the device

$$N_e = (F - 1)GN_{i,T_0} = (F - 1)GkT_0 B = GkT_{e,amp} B$$

$$T_{e,amp} = (F - 1)T_0$$

Below is only true when the noise power input is of $kT_0 B$, so not true in general

$$N_o = FGN_i \text{ only for } N_i = N_{i,T_0} = kT_0 B$$

$$N_{o,dB} = NF + G_{dB} + N_{i,dB} \text{ only for } N_i = N_{i,T_0} = kT_0 B$$

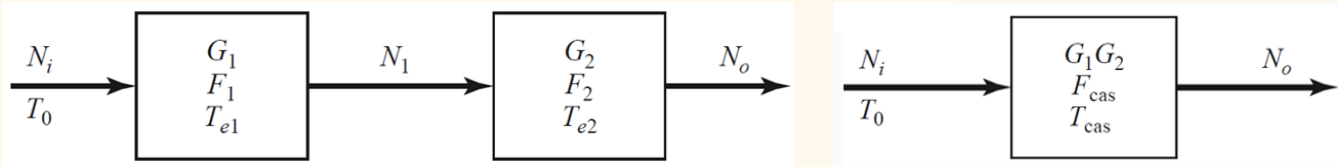
$$N_{o,dB} = NF + G_{dB} + 10\log(kT_0 B) \text{ only for } N_i = N_{i,T_0} = kT_0 B$$

All above is only for active devices, for passive attenuators of linear loss L (including lossy TL) in $T_{operating}$

$$F = 1 + \frac{(L-1)T_{operating}}{T_0} \quad \text{if } T_{operating} = T_0 \Rightarrow F = L, NF = L(dB)$$

Cascaded System (Friss Equation)

Noise Factor (F)/Noise Figure (NF) of a cascaded system is characterized when the input is $kT_0 B$



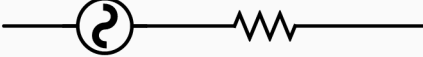
$$T_{e,cas} = T_{e,1} + \frac{T_{e,2}}{G_1} \quad F_{cas} = F_1 + \frac{F_2 - 1}{G_1} \quad T_{e,cas} = (F_{cas} - 1)T_0$$

For arbitrary number of stages

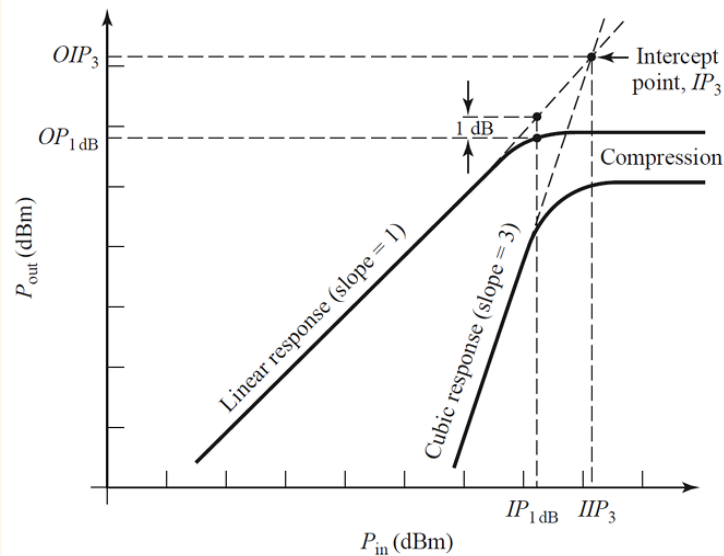
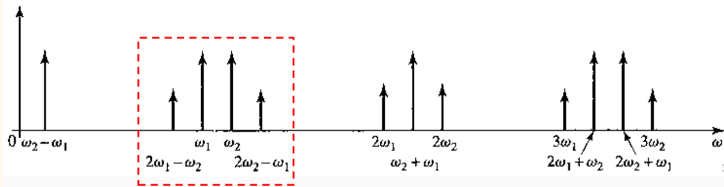
$$T_{e,cas} = T_{e,1} + \frac{T_{e,2}}{G_1} + \frac{T_{e,3}}{G_1 G_2} + \dots \quad F_{cas} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \dots$$

For $G_1 > 1$, the first stage should have low noise, since the cascaded NF depends heavily on it

NOTE: ALL ABOVE CALCULATION ARE IN LINEAR NOT dB

Thermal (Johnson) noise	
Resistor R at temperature T  (noisy resistor) + V_n - Noiseless Resistor R  rule of thumb: 50 ohm at room temperature over bandwidth of 1 Hz will get 1nV	$P_{thermal\ noise,\ ave,\ total\ available} = 4kTB$ $P_{thermal\ noise,\ ave,\ to\ a\ matched\ load} = kTB$ $V_{thermal\ noise,\ rms} = \sqrt{4kTBR} = 2\sqrt{kTBR}$ rms noise voltage across conductor of resistance R, due to thermal noise, obtained with $P = V^2/R$ $k = Boltzmann's\ constant = 1.380649 \times 10^{-23} J/K$ B (Hz) is single sided, only positive spectrum, with rectangular window
Shot Noise	
$I_{shotnoise,\ rms} = \sqrt{2qBI_{DC}}$ $q = elementary\ charge = 1.60217663 \times 10^{-19} C$ B (Hz) is single sided, only positive spectrum, with rectangular window	
Signal-to-noise ratio (SNR, S/N)	
$SNR = \frac{P_{signal}}{P_{noise}} \quad SNR(dB) = 10 \log_{10} \left(\frac{P_{signal}}{P_{noise}} \right) = 20 \log_{10} \left(\frac{V_{signal}}{V_{noise}} \right)$	
Noise Figure F	
$F = \frac{S/N_{input}}{S/N_{output}} = \frac{P_{SI}/P_{NI}}{P_{SO}/P_{NO}} = \frac{P_{SI}/P_{NI}}{\frac{G \times P_{SI}}{G \times (P_{NI} + P_{NA})}} = 1 + \frac{P_{NA}}{P_{NI}}$ If it's in dB, then the amount of noise added is F_{dB}, or equivalently, the amount of SNR drop is F_{dB}	

Nonlinearity



$$v_o = c_0 + c_1 v_i + c_2 v_i^2 + c_3 v_i^3$$

$$\text{where, } v_i = A \cos(\omega t)$$

$$= (c_0 + \frac{c_2 A^2}{2}) + (c_1 A + \frac{3c_3 A^3}{4}) \cos \omega t + \frac{c_2 A^2}{2} \cos 2\omega t + \frac{c_3 A^3}{4} \cos 3\omega t$$

Fundamental

Harmonics

$$y(t) \approx c_0 + c_1 x(t) + c_2 x(t)^2 + c_3 x(t)^3$$

$$\text{where, } x(t) = A(\cos(\omega_1 t) + \cos(\omega_2 t))$$

- A two-tone input voltage with two closely spaced frequencies,

- DC and fundamental components

$$\left(c_0 + c_2 A^2 \right) + \left(\left(c_1 A + \frac{9}{4} c_3 A^3 \right) (\cos \omega_1 t + \cos \omega_2 t) \right)$$

- Second and third harmonic terms

$$\left(\frac{c_2 A^2}{2} (\cos 2\omega_1 t + \cos 2\omega_2 t) \right) + \left(\frac{c_3 A^3}{4} (\cos 3\omega_1 t + \cos 3\omega_2 t) \right)$$

- Second-order inter-modulation(IM2) products

$$c_2 A^2 (\cos(\omega_1 + \omega_2) t + \cos(\omega_2 - \omega_1) t)$$

- Third-order inter-modulation(IM3) products

$$\frac{3}{4} c_3 A^3 (\cos(2\omega_1 + \omega_2) t + \cos(2\omega_1 - \omega_2) t + \cos(2\omega_2 + \omega_1) t + \cos(2\omega_2 - \omega_1) t)$$

$$OP_{1dB} = IP_{1dB} + G_{dB} - 1$$

$$G_{dB, \text{ no compression}} = 20 \log(c_1)$$

- Gain is defined to be at the fundamental component only
- Increase input power until the gain is 1dB less than ideal
- bigger better, bigger dynamic range
- 1 dB corresponds to 10% decrease in gain

$$A_{1dB} = \sqrt{\frac{c_1 (10^{-1/20} - 1)}{\frac{3}{4} c_3}} \quad (\text{volts})$$

NOTE: A is amplitude, need find rms for power

$$IP_{1dB} = 10 \log\left(\frac{\frac{1}{2} A_{1dB}^2}{Z_0}\right) \quad (dBW)$$

$$OIP_3 = IIP_3 + G_{dB}$$

- Use two tones similar in frequency, at low power (10dB less than IP_{1dB}) and slowly increase the input power for both
- With spectrum analyser with narrow RBW to reduce noise floor (allow small IMD to be seen), plot output vs input power of the fundamentals and IM3 (all 4, not just the two near passband)
- IM3 should have slope of 3, fundamental has slope of 1
- Extrapolate, the point of interception is at (IIP_3 , OIP_3)
- It's the input power at which the wanted signal and the 3rd order intermodulation products are equal, if there is no compression
- bigger better, 3rd order intermodulation products are lower than fundamental across bigger range of input power

$$A_{IIP_3} = \sqrt{\frac{4}{3} \left| \frac{c_1}{c_3} \right|} \quad (\text{volts})$$

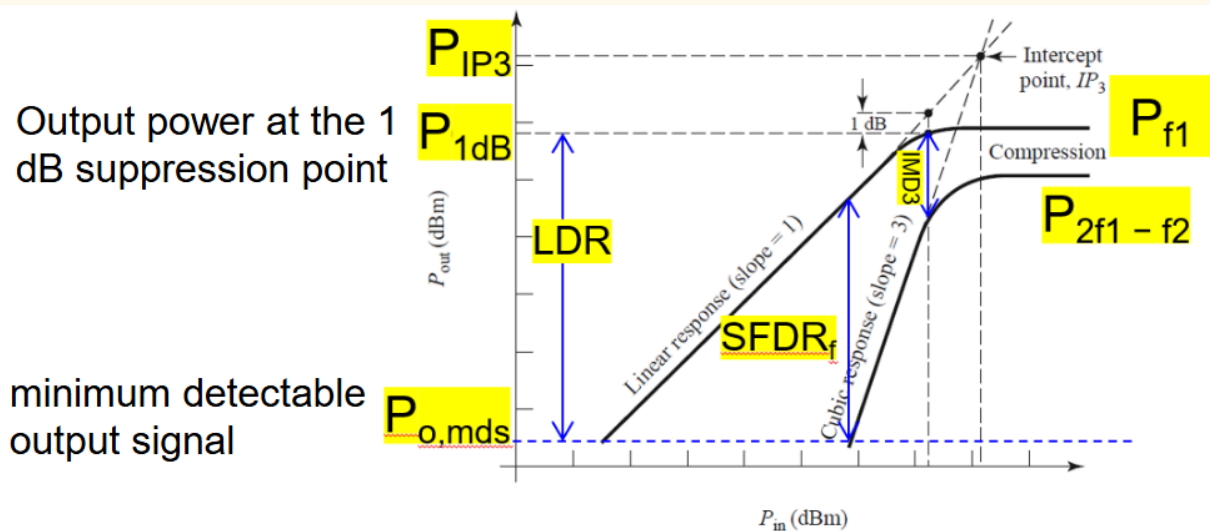
NOTE: A is amplitude, need find rms for power

$$IIP_3 = 10 \log\left(\frac{\frac{1}{2} A_{IIP_3}^2}{Z_0}\right) \quad (dBW)$$

P_o , and $P_o, 3$ being measurement of the output power

$$IMD_3 = P_o + P_{o,3} \quad (dB)$$

$$OIP_3 = P_o + \frac{IMD_3}{2} \quad (dBW)$$

$$IIP_3 = IP_{1dB} + 10 \text{ dB} \quad OIP_3 = OP_{1dB} + 10 \text{ dB}$$


Output power at the 3rd order intercept point (IP3) $P_{IP3} \text{ (dBm)} = P_{1dB} \text{ (dBm)} + 10 \text{ (dB)}$

spurious free
dynamic range

$$\text{SFDR}_f \text{ (dB)} = (P_{\text{IP3}} - P_{\text{o, mds}}) \times 2 / 3$$

- the power output MDS is taken as the output noise floor (calculated from N_i to N_o) (if minimum X dB above thermal noise is not specified)

Below are just estimation, as the phase of each stage is unknown, so the intermodulation product can be added in phase or cancel out (possible to design cancellation and increase the dynamic range)

$$\begin{aligned} (\text{OIP3}^T)^{-\frac{1}{2}} &\approx (G_m \dots G_2 \text{OIP3}_1)^{-\frac{1}{2}} + \dots (G_m \text{OIP3}_{(m-1)})^{-\frac{1}{2}} \\ &\quad + (\text{OIP3}_m)^{-\frac{1}{2}}. \end{aligned} \quad (7.12)$$
$$\left(\frac{1}{\text{IIP}3^{\text{T}}}\right)^{\frac{1}{2}} \approx \left(\frac{1}{\text{IIP}3_1}\right)^{\frac{1}{2}} + \dots + \left(\frac{G_{(m-2)} \dots G_1}{\text{IIP}3_{(m-1)}}\right)^{\frac{1}{2}} + \left(\frac{G_{(m-1)} \dots G_1}{\text{IIP}3_m}\right)^{\frac{1}{2}}. \quad (7.13)$$

- IIP3 of last stage dominates

Unorganized Cascade Intercept Method (voltage random phase and power adds, less conservative)

$$(\text{OIP3}^T)^{-1} \approx (G_m \dots G_2 \text{OIP3}_1)^{-1} + \dots + (G_2 \text{OIP3}_{(m-1)})^{-1} + (\text{OIP3}_m)^{-1} \quad (7.15)$$

and the total IIP3, IIP3^T , is obtained from (since $\text{IIP3} = \text{OIP3}/G$)

$$\left(\frac{1}{\text{IIP3}^T} \right) \approx \left(\frac{1}{\text{IIP3}_1} \right) + \left(\frac{G_1}{\text{IIP3}_2} \right) + \dots + \left(\frac{G_{(m-1)} \dots G_1}{\text{IIP3}_m} \right). \quad (7.16)$$

- IIP3 of last stage dominates

Cascaded $\text{OP}_{1\text{dB}}$ (from Steer Volume 4)

Below are just estimation, as the phase of each stage is unknown

$$(P_{1\text{dB}}^T)^{-\frac{1}{2}} \approx (G_2 P_{1,1\text{dB}})^{-\frac{1}{2}} + (P_{2,1\text{dB}})^{-\frac{1}{2}}. \quad (7.5)$$

Note that here the gains and the powers are absolute quantities and not in decibels. For example, $P_{1\text{dB}}^T$ is the power in watts at the 1 dB gain compression point.

If required and if a more realistic estimate is required and can be justified, then a less conservative formula that can be used to estimate the gain compression level of two cascaded stages is

$$(P_{1\text{dB}}^T)^{-1} \approx (G_2 P_{1,1\text{dB}})^{-1} + (P_{2,1\text{dB}})^{-1}. \quad (7.6)$$

It is usually better to use the more conservative estimate of compression in Equation (7.5).

- $P_{1\text{dB}}$ here refers to the output power $\text{OP}_{1\text{dB}}$
- $\text{OP}_{1\text{dB}}$ of last stage dominates